

Managing Project Time

Assoc. Professor Darwin Kim Bulawan



Course Map

Week Number	Inclusive Dates	Module Number	Assessments	Topics
1	Sep 6	1	Quiz	Course Introduction, Discussion of Deliverables, Introduction to IT Project Management, and Class Consultations and Submission of Deliverables
2-3	Sep 13 - 20	2	Quiz	Managing Project Scope, and Class Consultations and Submission of Deliverables
4-5	Sep 27, Oct 4	3	Quiz	Managing Project Time and Class Consultations and Submission of Deliverables
6-7	Oct 11 -18	4	Quiz	Managing Project Cost and Class Consultations and Submission of Deliverables
8	Oct 25	-	Exam	Midterm Examination
9	Nov 1	-		
10-11	Nov 8 - 15	5	Quiz	Managing Project Leadership and Communication Management, and Class Consultations and Submission of Deliverables
12-13	Nov 22 - 29	6	Quiz	Managing Project Risk, and Class Consultations and Submission of Deliverables
14-15	Dec 6 - 13	7	Quiz	Managing Project Procurement and Class Consultations and Submission of Deliverables
16-18	Dec 20 - Jan 3	-	-	Christmas Break
19	Jan 10	-	Report, Presentation	Class Presentations, and Class Consultations and Submission of Deliverables
20	Jan 17	-	Exam	Final Examination



Learning Objectives

- Discuss the importance and processes of project time management
- Describe how IT project managers use network diagrams and dependencies to assist in activity sequencing
- Explain how various tools and techniques help project managers perform activity duration estimates
- Discuss Gantt chart for planning and controlling project schedule
- Describe how the Program Evaluation and Review Technique (PERT) affect schedule development

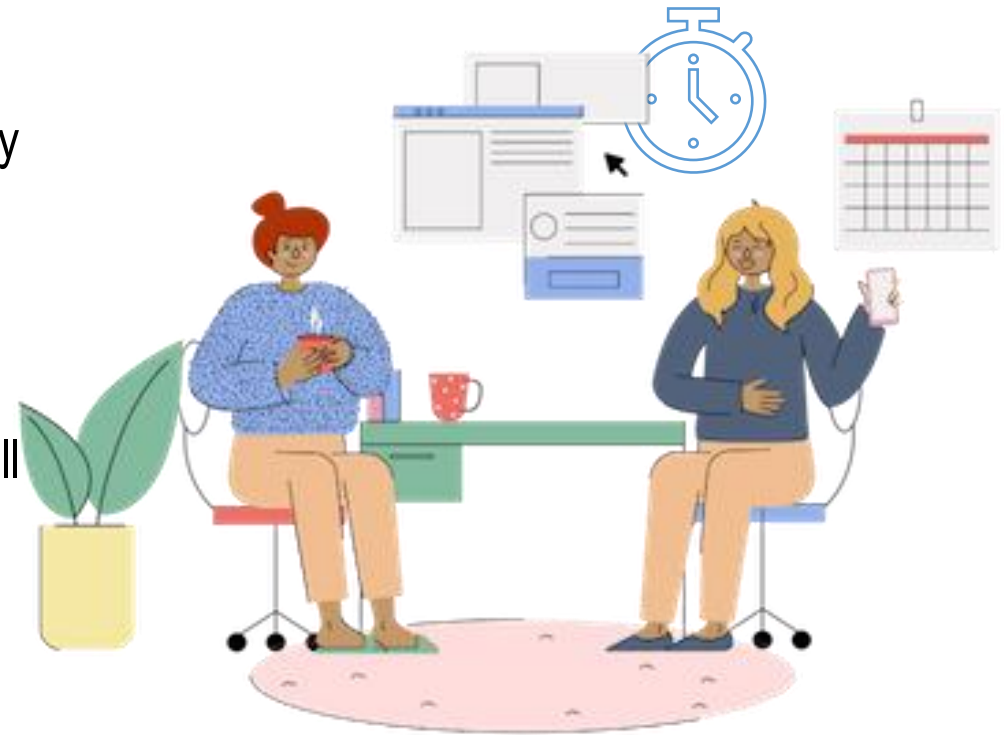


PROJECT TIME/SCHEDULE MANAGEMENT

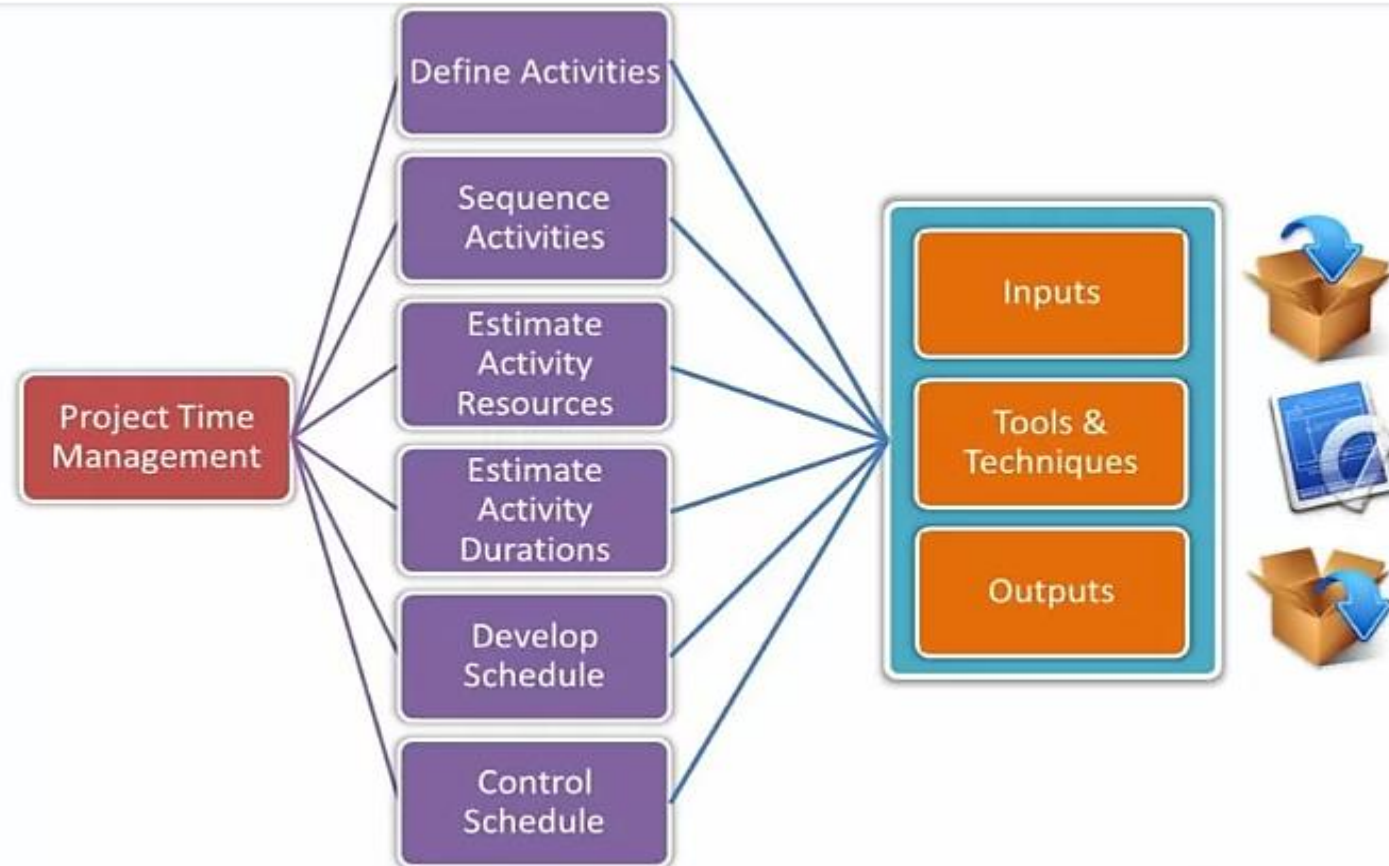
- 🕒 **Project Time Management** includes the processes required to manage timely completion of the project.
- 🕒 discipline of planning and scheduling one's activities so that they fit in with other commitments, resources, and goals.

KEY BENEFIT:

provides guidance and direction on how the project schedule will be managed throughout the project.



PROJECT TIME/SCHEDULE MANAGEMENT

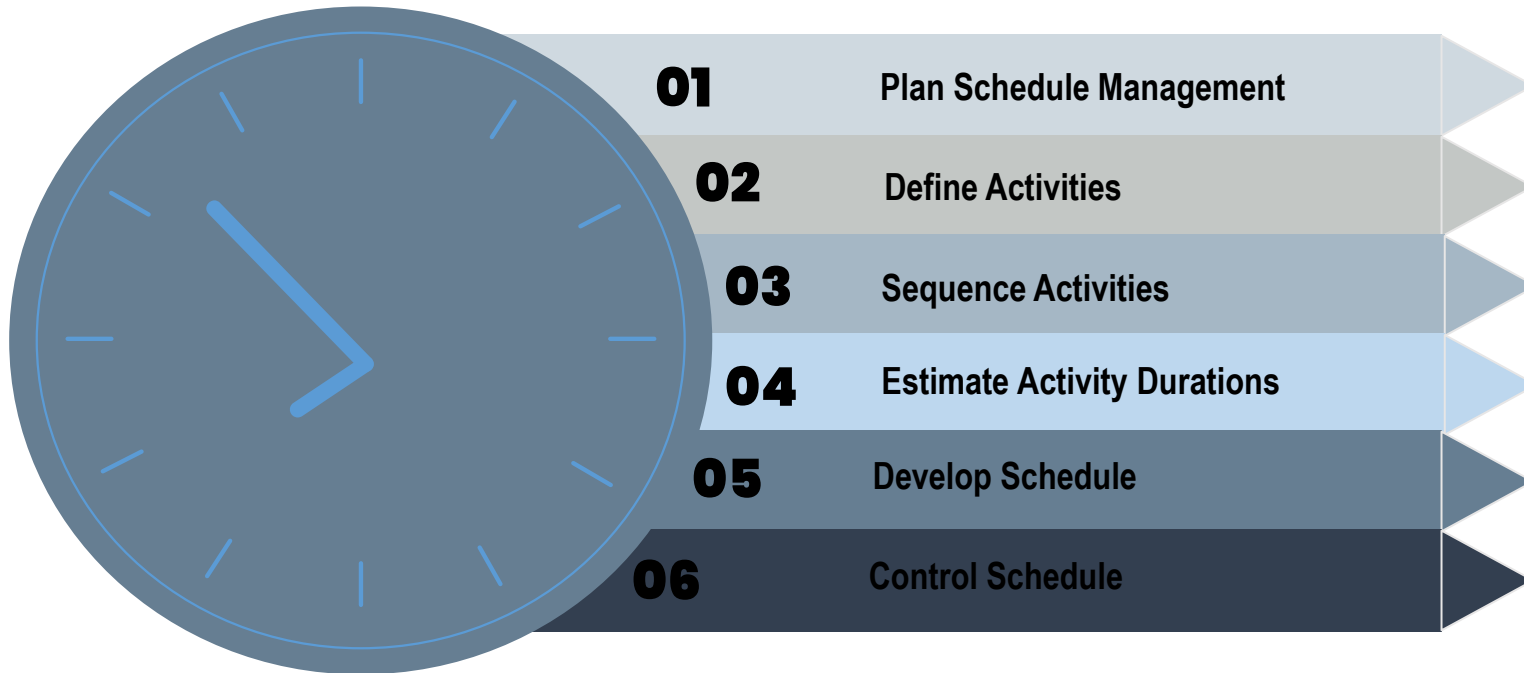


TIME MANAGER

- ☐ Have well-defined goals for the year and know how to work towards them.
- ☐ Can track their progress towards their goals so that they can adjust their plans accordingly.
- ☐ Can manage time through various techniques like setting a daily schedule, using a calendar as a reminder tool, using mobile apps to track your daily schedule, etc.



PROJECT SCHEDULE MANAGEMENT



PLAN SCHEDULE MANAGEMENT

- 🕒 The process of establishing the policies, procedures, and documentation for planning, developing, managing, executing, and controlling the project schedule.
- 🕒 **Key benefit:** is to provide guidance and direction on how the project schedule will be managed.



SCHEDULING TOOLS OR TECHNIQUES

Commonly Used Methods are:



GANTT CHART



NETWORK DIAGRAM (PERT/CPM)



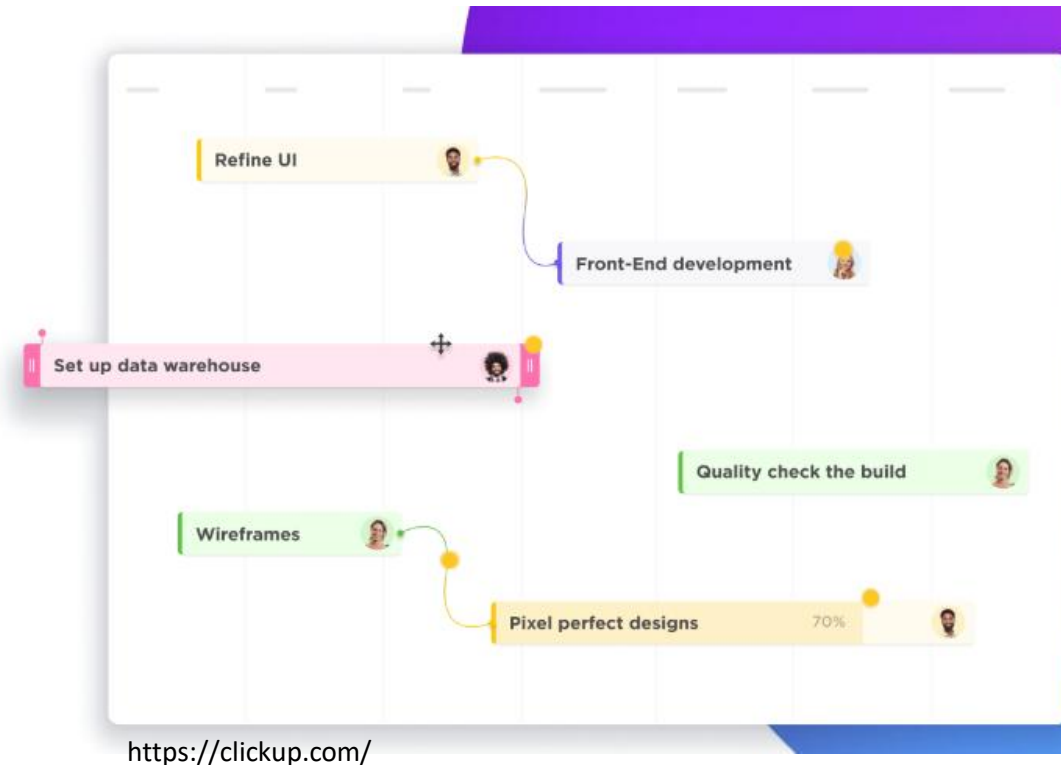
SCHEDULING TOOLS

Gantt Chart

A bar chart that is a visual representation of the **sequencing** and **duration of activities** on any given project.

Useful:

- ✓ Easy to read
- ✓ Give each team members overview of the project.
- ✓ Indicate clearly the status of each activity
- ✓ Can be drawn to show the budgets, equipment's usage



GANTT CHART

 **Gantt Chart-** are used as a tool to monitor and control the project progress.

- Graphical presentation that displays activities:
 - Time- measured on the horizontal axis.
 - Horizontal bar- drawn proportionately to an activity's expected completion time.
 - Each activity is listed on the vertical axis.



APPLICATION OF GANTT CHART

-  **Gantt Chart-** used as a visual aid for tracking the progress of project activities.
-  Appropriate percentage of a bar is shaded to document the completed work.
-  Time manager easily see if the project is progressing on schedule with respect to the earliest possible completion times.



NETWORK DIAGRAM

- ✚ Is a graphical depiction of project tasks and their inter-relationships.
- ✚ Is ordering of tasks is shown by connection with predecessor and successor tasks
- ✚ Is a critical path scheduling technique used for controlling resources.

✚ **Critical path scheduling**

- A scheduling technique whose order and duration of a sequence of task activities directly affects the completion date of a project.



PROGRAM EVALUATION REVIEW TECHNIQUE (PERT)

PERT is a techniques used to calculate the **Expected Time** for a task.

PERT is a technique that uses:

- Optimistic time (O),
- Pessimistic time (P), and
- Realistic Time (R),

estimates to calculate the EXPECTED TIME (ET) or a particular task.



PROGRAM EVALUATION REVIEW TECHNIQUE (PERT)

PERT is a techniques used to calculate the **Expected Time** for a task.

PERT is a technique that uses:

- **Optimistic time (O) & Pessimistic time (P)**- reflect the minimum, and maximum possible periods of the time for an activity to be completed
- **Realistic Time (R)**- reflect the project manager's " Best Guess" of the amount of the time required for a task completion

estimates to calculate the EXPECTED TIME (ET) or a particular task.



PROGRAM EVALUATION REVIEW TECHNIQUE (PERT)

CALCULATING EXPECTED COMPLETION TIME (ET)

$$ET = \frac{o + 4r + p}{6}$$

Because the expected Completion time should be closer to the realistic time (r), it is typically weighed Four times more than the Optimistic time (o) and the Pessimistic time (p). Once you add these values together , it must be divided by 6 to determine the Expected Time for a task.



DEFINE ACTIVITIES

- 🕒 The process of identifying and documenting the specific actions to be performed to produce the project deliverables.
- 🕒 **Key benefit:** is to break down work packages into activities that provide a basis for estimating, scheduling, executing, monitoring, and controlling the project work..



SEQUENCE ACTIVITIES

- ⌚ The process of identifying and documenting relationships among the project activities. Finding Lead or Lag time between activities should be considered to support a realistic schedule. This can be performed by software or manually.
- ⌚ **Key benefit:** process is that it defines the logical sequence of work to obtain the greatest efficiency given all project constraint.



ESTIMATE ACTIVITY RESOURCES

- ⌚ This is done to Estimate the type and quantities of materials, human resources and equipment required to perform each activity. .
- ⌚ **Key benefit:** Process is that it identifies the type, quality and characteristics of resources required to complete the activity which allows more accurate cost and duration estimate..



ESTIMATE ACTIVITY DURATION

- ⌚ This process of approximating the number of work periods needed to complete individual activities with estimated resources.
- ⌚ **Key benefit:** this process is that it provide the amount of the time each activity will take to complete, which is a major input to Develop schedule process.



DEVELOP SCHEDULE

- 🕒 The process of analyzing activity sequences, durations, resource requirements, and schedule constraints to create the project schedule model for project execution and monitoring and controlling.
- 🕒 **Key benefit:** is to break down work packages into activities that provide a basis for estimating, scheduling, executing, monitoring, and controlling the project work..



CONTROL SCHEDULE

 The process of monitoring the status of the project to update the project schedule and manage changes to the schedule baseline.

This is also concerned with:

- ✓ Determine current status of the project activities
- ✓ Influence the factors that cause changes or delay
- ✓ Manage actual changes as they occur to reduce the risk

 **Key benefit:** this process is that it provides the means to recognize deviation from the plan and take corrective action and preventive actions and thus minimizes risk.



Module 3 Activity

(Individual Activity) (10 points)

Link: <https://forms.gle/MRoXpNsm5JTgh2xM8>

Further instructions will be announced via Google Forms.

Due Date: **18 October**

